

Contribution Record — Smart Scientific Calculator

Repository: <https://github.com/Krishnarajss/Smart-Scientific-Calculator>

1. Team Members

#	Name	Registration No.	GitHub Username	Role / Responsibility
1	Siddhartha Tiwari	20243275	princetiwari1204	Team Leader — Full-stack implementation (all iterations)
2	Krishnaraj Singh	20243278	Krishnarajss	SRS, Architecture, UML Diagrams, Use Cases, Repo Setup
3	Sourabh Verma	20243279	sourabh-verma18	Testing (Test Cases & Evidence), Project Management Docs
4	Umar Rashid	20243591	umarrasheed77	Demo Video, Mid-Semester Project Report

2. Individual Contribution Summary

Siddhartha Tiwari (20243275) — Team Leader

Implemented the complete source code across all six iterations:

- Iteration 1: Basic arithmetic operations (+, -, *, /, %, AC, DEL)
- Iteration 2 & 3: Scientific functions ($\sqrt{}$, sin, cos, x^y)
- Iteration 4: Error handling (division by zero, invalid domain, syntax errors)
- Iteration 5: Responsive grid-based UI layout, real-time display, button feedback
- Iteration 6: Node.js/Express backend server, MongoDB integration via Mongoose, history API
- Authored frontend files: public/index.html, public/script.js, public/style.css
- Authored backend: server.js; configured dependencies in package.json and package-lock.json
- Extended Readme.md with iteration-wise project documentation and directory structure
- Created PR #1 (princetiwari → main) adding the full project structure, and merged it into main

Krishnaraj Singh (20243278)

- Created and owns the GitHub repository (Krishnarajss/Smart-Scientific-Calculator)
- Added the initial project description to Readme.md
- Prepared and uploaded all requirements and design artifacts to the SRS branch:
- SRS document.pdf — Software Requirements Specification
- Architecture_Diagram_Clean.pdf — System architecture diagram
- UML_Diagrams.pdf — UML class/sequence diagrams
- Use_Cases_or_User_Stories.pdf — Use-case descriptions and user stories
- Maintained the SRS branch containing all requirements and design documents

Sourabh Verma (20243279)

- Designed and documented 19 test cases covering all six iterations:
- Basic arithmetic (TC-001 to TC-006)
- Scientific functions (TC-007 to TC-010)
- Error handling (TC-011 to TC-013)
- UI and interaction (TC-014 to TC-015)
- Database history integration (TC-016 to TC-019)

- Prepared test evidence documentation (execution results and screenshots)
- Prepared all Project Management documents on the project-management-and-testing branch:
- Project_Plan.pdf — Timeline (Day 1-Day 7), milestones, task allocation, risks and remaining work
- Meeting_Log_Final.pdf — Three meeting records with agendas, decisions and action items
- Contribution_Record.pdf — This document

Umar Rashid (20243591)

- Recorded, edited and produced the demonstration video (demo_video.mp4), covering:
- Project title and team introduction
- Problem explanation and proposed solution overview
- Architecture walkthrough
- Live demo of working features
- Current limitations and remaining work
- Individual contribution statements
- Authored the Mid-Semester Project Report (Smart_Scientific_Calculator_SEPM_Report_with_Intro.pdf):
- Cover page, introduction, problem statement, objectives, scope
- Requirements summary, system design overview
- Implementation progress with screenshots
- Testing summary, project management status
- Challenges, future work, and references

3. Contribution Percentage

Member	Reg. No.	Contribution (%)
Siddhartha Tiwari	20243275	25%
Krishnaraj Singh	20243278	25%
Sourabh Verma	20243279	25%
Umar Rashid	20243591	25%
Total		100%

4. Git Commit History

The following is the actual commit history extracted from the GitHub repository (19 commits across 5 branches; the 6 clean-up deletions are shown as one row). All times are IST. Commit dates show when files were uploaded to GitHub, not when the work was carried out.

main Branch

#	SHA	Author	Date	Message
1	6a071bf	Krishnaraj Singh	2 Oct 2026, 12:57 PM	Add initial project description to Readme.md
2	27cbe28	Siddhartha Tiwari	2 Oct 2026, 5:55 PM	Add full project structure across all iterations
3	cff6fdc	Siddhartha Tiwari	2 Oct 2026, 6:16 PM	Merge pull request #1 from Krishnarajss/princetiwari

SRS Branch

#	SHA	Author	Date	Message
4	2e9a152	Krishnaraj Singh	2 Oct 2026, 2:57 PM	Add files via upload (SRS document.pdf)
5	1394bd2	Krishnaraj Singh	2 Oct 2026, 3:05 PM	Add files via upload (Use_Cases_or_User_Stories.pdf)
6	8d0ea6c	Krishnaraj Singh	2 Oct 2026, 3:15 PM	Add files via upload (Architecture_Diagram_Clean.pdf)
7	8da4bfa	Krishnaraj Singh	2 Oct 2026, 3:19 PM	Add files via upload (UML_Diagrams.pdf)

project-management-and-testing Branch

#	SHA	Author	Date	Message
8	d15e31d	Sourabh Verma	2 Oct 2026, 10:02 PM	Add files via upload (Contribution_Record.pdf, Meeting_Log.pdf, Project_Plan.pdf)
9	a7e8aec - 71f40f3	Sourabh Verma	2 Oct 2026, 10:07 PM	Clean-up: delete copied code files (node_modules, public, Readme.md, package-lock.json, server.js, package.json) so the branch holds only project-management documents (6 commits)
10	9e6ac2a	Sourabh Verma	3 Oct 2026, 10:44 PM	Add files via upload (Project_Plan.pdf updated)
11	3c38ad9	Sourabh Verma	3 Oct 2026, 10:57 PM	Delete Meeting_Log.pdf
12	2989c48	Sourabh Verma	3 Oct 2026, 10:57 PM	Add files via upload (Meeting_Log_Final.pdf)

REPORT-AND-DEMO Branch

#	SHA	Author	Date	Message
13	8ea2429	Umar Rashid	3 Oct 2026, 10:44 PM	Add files via upload (Smart_Scientific_Calculator_SEPM_Report_with_Intro.pdf)
14	1414b71	Sourabh Verma	3 Oct 2026, 11:24 PM	Add files via upload (demo_video.mp4) *

* The demo video (demo_video.mp4) was recorded and produced by Umar Rashid; the file was uploaded to the repository from a team member's GitHub account. The princetiwari branch contains commit 27cbe28 (listed under main), which was merged through PR #1.

Contributor Statistics (from GitHub)

Contributor	GitHub Username	Commits	Files Changed
Siddhartha Tiwari	princetiwari1204	2	public/, node_modules/, Readme.md, package.json, package-lock.json, server.js
Krishnaraj Singh	Krishnarajss	5	Readme.md, SRS document.pdf, Architecture_Diagram_Clean.pdf, UML_Diagrams.pdf, Use_Cases_or_User_Stories.pdf
Sourabh Verma	sourabh-verma18	11 (incl. 1 upload of Umar Rashid's demo video)	Project_Plan.pdf, Meeting_Log_Final.pdf, Contribution_Record.pdf (project-management-and-testing branch)

Contributor	GitHub Username	Commits	Files Changed
Umar Rashid	umarrasheed77	1 (+ demo video, uploaded via another member's account)	Smart_Scientific_Calculator_SEPM_Report_with_Intro.pdf (REPORT-AND-DEMO branch); demo_video.mp4

5. Task Allocation Table

Task	Assigned To	Status
Repository creation and setup	Krishnaraj Singh	Completed
SRS document	Krishnaraj Singh	Completed
Use-case diagram / User stories	Krishnaraj Singh	Completed
Architecture diagram	Krishnaraj Singh	Completed
UML diagrams	Krishnaraj Singh	Completed
Iteration 1: Basic arithmetic implementation	Siddhartha Tiwari	Completed
Iteration 2 & 3: Scientific functions	Siddhartha Tiwari	Completed
Iteration 4: Error handling	Siddhartha Tiwari	Completed
Iteration 5: UI design and interaction	Siddhartha Tiwari	Completed
Iteration 6: Backend server + MongoDB	Siddhartha Tiwari	Completed
Test-case design and execution (19 cases)	Sourabh Verma	Completed
Test evidence documentation	Sourabh Verma	Completed
Project plan	Sourabh Verma	Completed
Meeting log	Sourabh Verma	Completed
Contribution record	Sourabh Verma	Completed
Mid-semester project report	Umar Rashid	Completed
Demonstration video	Umar Rashid	Completed
Final ZIP packaging and submission	All members	In Progress

6. Daily Progress Record (1-Week Schedule)

Day	Tasks Completed	Members Involved
Day 1	Kick-off meeting, topic selection, problem statement, SRS drafting, architecture design; Iteration 1 (basic arithmetic)	All members; Krishnaraj Singh; Siddhartha Tiwari
Day 2	Iterations 2 & 3 (scientific functions); use-case diagram and user stories	Siddhartha Tiwari; Krishnaraj Singh
Day 3	Iteration 4 (error handling)	Siddhartha Tiwari
Day 4	Iteration 5 (responsive UI); progress-review meeting; Iteration 6 started	Siddhartha Tiwari; All members
Day 5	Iteration 6 (Express + MongoDB history API); UML diagrams; test-case design (19 cases)	Siddhartha Tiwari; Krishnaraj Singh; Sourabh Verma
Day 6	Full testing and evidence; project plan and meeting log; final integration meeting; code, SRS and design files uploaded to GitHub	Sourabh Verma; All members; Siddhartha Tiwari; Krishnaraj Singh
Day 7	Mid-semester report, demonstration video, repository upload and final ZIP packaging	Umar Rashid; All members

7. Individual Contribution Declaration

By listing our names below, each team member confirms that the contributions described in this document are accurate and that all members participated meaningfully in the project.

Member	Reg. No.	Signature / Confirmation	Date
Siddhartha Tiwari	20243275		
Krishnaraj Singh	20243278		
Sourabh Verma	20243279		
Umar Rashid	20243591		